

Irfan Ullah

✉ irfan@tamu.edu [in](#) [LinkedIn](#) [globe](#) [Research Portfolio](#) [📍](#) Bryan, TX — Open to Relocation

Professional Summary

Power systems engineer and Ph.D. candidate at Texas A&M University with 5+ years of experience in power-system analysis, distribution-system modeling, DER integration, and resilient grid operation. Experienced with OpenDSS, PowerModelsONM, PSCAD, MATLAB/Simulink, Python, and Julia for power-system modeling, simulation automation, and distribution-system analysis. Research focuses on black start restoration, microgrid reconfiguration, grid-forming inverter constraints, voltage regulation, and large-scale simulation studies for active distribution networks. Proven ability to develop operational strategies for resilient electric infrastructure through research collaborations with Los Alamos National Laboratory and DOE-sponsored projects.

Experience

Texas A&M University, Electrical & Computer Engineering

Aug 2022 – Present

Graduate Research Assistant – Power & Energy Group

College Station, TX

- Conduct distribution system studies on large-scale utility feeders supporting critical power systems, evaluating restoration strategies, voltage stability, and thermal loading under renewable penetration and load conditions.
- Developed optimization-based black start restoration strategies for utility-scale distribution systems using mixed-integer programming and grid-forming inverter operational constraints.
- Performed detailed distribution system studies on the Iowa 240-node and Smart-DS feeders to evaluate voltage stability, power flow feasibility, and DER integration impacts.
- Developed automated evaluation workflows for power-system restoration studies, comparing system-model features, mathematical formulations, and solution strategies across large-scale distribution networks to assess restoration feasibility, DER coordination, voltage regulation, topology decisions, and computational performance.
- Mentor undergraduate researchers on distribution system modeling, simulation setup, and automation of power system simulation workflows while coordinating restoration studies and research activities with Purdue University.
- Prepared technical reports and analysis for DOE-sponsored research programs focused on resilient operation of modern distribution power systems.

Los Alamos National Laboratory

Summers 2024 & 2025

Summer Research Intern – Power Systems

Los Alamos, NM

- Collaborated with Los Alamos National Laboratory scientists to analyze operational strategies for resilient distribution power systems using large-scale simulation and optimization models.
- Developed uncertainty-aware load-block modeling approaches for distribution networks to evaluate load variability and improve reliability of restoration and microgrid reconfiguration strategies.
- Developed automated Python workflows integrating OpenDSS, Julia-based PowerModelsDistribution and PowerModelsONM optimization tools, PowerPlots, and Excel-based data processing to run large-batch distribution-system simulations, visualize system results, extract performance metrics, compare scenarios, and accelerate engineering analysis.
- Enhanced distribution grid simulation capabilities in PowerModelsONM to represent voltage regulation, cold load pickup behavior, and inverter-based distributed energy resources.
- Diagnosed and corrected power flow and voltage constraint violations in PowerModelsONM, improving restoration feasibility in a DOE grid resilience platform recognized with an R&D 100 Award.

Education

Texas A&M University

2022 – 2026 (expected)

Ph.D. in Electrical Engineering

College Station, TX

Relevant Coursework: Power System Analysis and Control, Optimization, Machine Learning

COMSATS University, Islamabad

2018 – 2020

M.S. in Electrical Engineering

Islamabad, PK

Relevant Coursework: Smart Grid, Power System Protection, Power System Analysis, Renewable Energy Technology

Thesis: Unified Power Quality Conditioner Based Power Quality Improvement in Microgrid

University of Engineering and Technology

2012 – 2016

B.S. in Electronics Engineering

Peshawar, PK

Final Year Project: Antenna Design for Wi-Fi Application

Technical Skills

Power Systems Analysis: Power-system modeling and power-flow analysis, distribution-system modeling, multi-phase unbalanced systems, voltage regulation, feeder reconfiguration, DER integration, microgrid operation, and grid restoration analysis.

Power System Tools: OpenDSS, PowerModelsDistribution, PowerModelsONM, PowerWorld, PSCAD, MATLAB/Simulink, PSS/E, CYME, ETAP.

Programming: Python (simulation automation and data analysis), Julia (JuMP optimization modeling), MATLAB scripting, Git/GitHub.

Optimization Methods: Mixed-Integer Linear Programming (MILP), robust optimization, rolling horizon optimization, convex relaxation, uncertainty modeling.

Engineering Tools: PSpice, Multisim, AutoCAD.

Publications (Selected)

I. Ullah, K. L. Butler-Purry, H. Nagarajan, "Two-Stage Robust Optimization Maximal Load Delivery (MLD) Framework on a Large-Scale Distribution System," *accepted in IEEE PES General Meeting (PESGM)*, 2026.

I. Ullah, KL Butler-Purry, R Bent, H Nagarajan, A Barnes, "Studying a Sequential Black Start Restoration Method Using PowerModelsONM.jl" *North American Power Symposium (NAPS)*, 2025.

W. Ahmad, L. Khan, Q. Khan and I. Ullah, "Advanced Sliding Mode Control Techniques Employed on Electric Hybrid Vehicle," *2024 IEEE Texas Power and Energy Conference (TPEC)*, College Station, TX, USA, 2024, pp. 1-6.

I. Ullah, S. Rahman, I. A. Khan, "SST-Based Marine Shipboard System for Improved Performance and Renewable Integration," *IEEE Texas Power and Energy Conference (TPEC)*, 2023.

Projects

- **Distribution System Resilience and DER Integration:** Performed restoration, power-loss, and DER-integration studies for distribution networks under high-renewable and extreme operating conditions.
- **SST-Based Power Systems and Data Center Energy Routing:** Analyzed SST-based shipboard power systems and UPQC applications for power-quality improvement, with ongoing research on multi-port SST energy-router architectures for converter-dominated data centers, storage integration, and bidirectional power routing.
- **Multi-Agent Grid Restoration Using PV-Based EV Stations:** Investigated a multi-agent framework for grid restoration in active distribution networks using PV-integrated EV charging stations.
- **Power Grid Cyber-Attack Detection Using Machine Learning:** Developed a course project for cyber-attack detection using autoencoder-based feature extraction and Random Forest classification.
- **Reliability Analysis of Smart Grids Under Cyber Attacks:** Evaluated smart-grid reliability under cyber-attack scenarios using Monte Carlo simulation and applied black start restoration for system recovery analysis.

Leadership and Professional Service

Communication Chair, Texas Power & Energy Conference (TPEC) (2026): Led technical communications and outreach for the Texas Power and Energy Conference, coordinating newsletters, sponsor communication, and industry speaker engagement.

Promotional Chair, TPEC (2024-2025): Supported conference growth through targeted outreach and coordination of promotional activities.

Technical Reviewer: IEEE Power & Energy Society General Meeting and TPEC (2023-2026), reviewing manuscripts on power systems, optimization, and distribution grid modeling.

Honors & Awards

- Selected for competitive research collaboration with Los Alamos National Laboratory staff scientists on large-scale optimization and resilience of complex engineering systems.
- Prime Minister National ICT Scholarship, fully funded national award recognizing academic excellence and leadership potential (2012–2016).
- Recipient of National ICT Initiative Research Grant for Wi-Fi antenna design, analysis, and prototyping (2015–2016).